1.5 Conditional Statements

Every theorem in geometry is really a conditional statement in disguise, so learning to dissect "if p, then q" into its converse, inverse, and contrapositive is what makes later proofs readable. Beyond just rewriting a sentence, this section asks you to judge *truth* — proving a statement false with a single counterexample, or recognizing when a conditional and its contrapositive must share a truth value while the converse might not. It also introduces the two rules that power two-column proofs: the Law of Detachment and the Law of Syllogism, along with the classic converse-error trap of "affirming the consequent."

Examples

Example 1: Hypothesis, conclusion, and truth value

Consider: "If a figure is a square, then it is a rectangle." Identify the hypothesis and conclusion, and determine whether the conditional is true.

Solution

Split the sentence at "if" and "then": hypothesis p: a figure is a square; conclusion q: it is a rectangle.

Check truth: every square has four right angles, which is exactly the definition of a rectangle, so whenever p holds, q holds too.

Answer: p: square; q: rectangle; the conditional is **true**.

Example 2: Converse, and why it can fail

Write the converse of "If a figure is a square, then it is a rectangle," and determine whether the converse is true.

Solution

Form the converse by swapping hypothesis and conclusion: "If a figure is a rectangle, then it is a square."

Test with a counterexample: a 4×8 rectangle has four right angles (satisfying the hypothesis) but unequal side lengths, so it is not a square.

Answer: Converse: "If a figure is a rectangle, then it is a square" — **false** (counterexample: any non-square rectangle).

Example 3: Inverse and why it can differ from the original

Write the inverse of "If it rains, then the ground is wet," and explain why a conditional and its inverse don't have to share a truth value.

Solution

Form the inverse by negating both parts: "If it does not rain, then the ground is not wet."

This can fail even when the original is true: a sprinkler could wet the ground with no rain, so "no rain" does not guarantee "ground not wet."

Answer: Inverse: "If it does not rain, then the ground is not wet" — not guaranteed true, even though the original conditional is.

Example 4: Contrapositive is always equivalent

Write the contrapositive of "If two lines are parallel, then they do not intersect," and explain why a contrapositive always shares the truth value of the original.

Solution

Form the contrapositive by negating both parts *and* swapping their order: "If two lines intersect, then they are not parallel."

Logically, $p \to q$ and $\neg q \to \neg p$ describe the same relationship from two angles — whenever one is true the other must be true (and likewise false), unlike the converse or inverse.

Answer: "If two lines intersect, then they are not parallel" — logically equivalent to the original.

Example 5: Disproving with a counterexample

Conditional: "If $x^2 = 9$, then $x = 3$." Show the statement is false by giving a counterexample.

Solution

To disprove a conditional, you need only one case where the hypothesis is true but the conclusion is false.

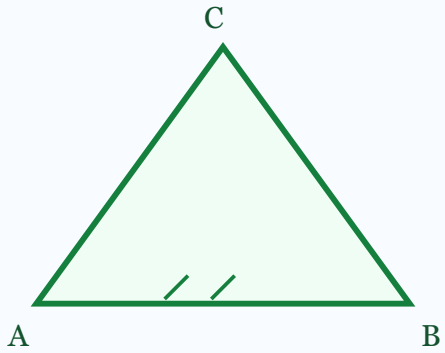
Let $x = -3$: then $x^2 = (-3)^2 = 9$, so the hypothesis holds — but $x = -3 \neq 3$, so the conclusion fails.

Answer: $x = -3$ is a counterexample, so the conditional as stated is false.

Example 6: Building a biconditional

Given two true conditionals — (1) If a triangle is equilateral, then all three sides are congruent; (2) If all three sides of a triangle are congruent, then it is equilateral — combine them into a biconditional statement.

Solution



Triangle ABC with all sides congruent — the two directions describe the same fact.

A biconditional joins $p \rightarrow q$ with its converse $q \rightarrow p$ using "if and only if," and is true exactly when *both* directions are true.

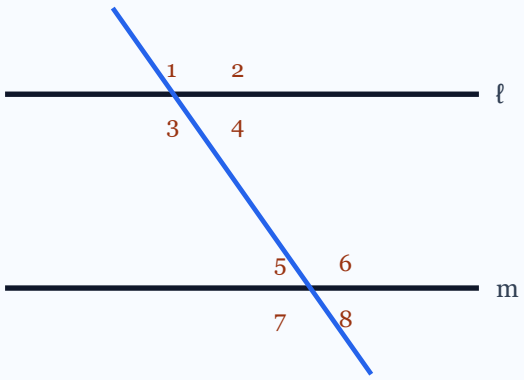
Both given conditionals are true, so they may be merged.

Answer: "A triangle is equilateral if and only if all three of its sides are congruent."

Example 7: Law of Detachment

Given: (1) If two lines are parallel, then their corresponding angles are congruent. (2) Lines ℓ and m are parallel. What can you validly conclude, and which law justifies it?

Solution



Lines ℓ and m are parallel, cut by a transversal.

Identify the forms: statement (1) is $p \to q$ (parallel ⇒ congruent corresponding angles); statement (2) asserts p is true.

Law of Detachment: if $p \to q$ is true and p is true, then q must be true.

Answer: The corresponding angles formed by ℓ, m, and the transversal are congruent (by the Law of Detachment).

Example 8: Law of Syllogism

Given: (1) If a figure is a square, then it is a rectangle. (2) If a figure is a rectangle, then it is a parallelogram. Combine these into one conditional, and name the law used.

Solution

Identify the chain: (1) is $p \to q$ (square ⇒ rectangle); (2) is $q \to r$ (rectangle ⇒ parallelogram) — the conclusion of (1) matches the hypothesis of (2).

Law of Syllogism: from $p \to q$ and $q \to r$, conclude $p \to r$.

Answer: "If a figure is a square, then it is a parallelogram" (by the Law of Syllogism).

Example 9: Spotting an invalid argument

Given: (1) If a quadrilateral is a square, then it has four right angles. (2) Quadrilateral R has four right angles. A student concludes R must be a square. Is this reasoning valid?

Solution

Compare to the Law of Detachment: that law requires the *hypothesis* (square) to be known true in order to conclude the *conclusion* (four right angles). Here the conclusion is given and the hypothesis is being guessed — that is not a recognized law.

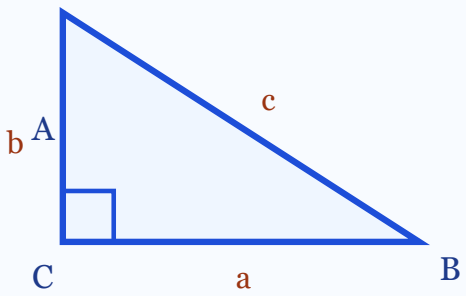
Counterexample exposing the flaw: a non-square rectangle also has four right angles, satisfying (2) without being a square.

Answer: Not valid — this is the converse error ("affirming the consequent"); R could be a rectangle instead of a square.

Example 10: Symbolic biconditional

Let p: "an angle is right" and q: "the angle measures $90°$." Write the biconditional in symbols and in words, and state its truth value.

Solution



The angle at C is right — exactly 90°, by definition.

Symbolic form: $p \leftrightarrow q$.

In words: "An angle is right if and only if it measures $90°$."

Both directions (right $\Rightarrow 90°$, and $90° \Rightarrow$ right) hold by definition, so the biconditional is true.

Answer: $p \leftrightarrow q$; the statement is **true**.

Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Hypothesis

"If n is even, then n is divisible by 2." Hypothesis?

Answer: **Verification:**

Problem 2: Conclusion

Same statement. Conclusion?

Answer: **Verification:**

Problem 3: Converse

If a polygon is a triangle, then it has 3 sides. Converse?

Answer: **Verification:**

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Problem 4: Inverse

If an angle is right, then it measures $90°$. Inverse?

Answer: **Verification:**

Problem 5: Contrapositive

If two lines are parallel, then they do not intersect. Contrapositive?

Answer: **Verification:**

Problem 6: Equivalent form

Which is logically equivalent to a conditional?

Answer: **Verification:**

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Problem 7: Counterexample

If $x^2 = 9$, then $x = 3$. Counterexample?

Answer: **Verification:**

Problem 8: Biconditional

When is a biconditional true?

Answer: **Verification:**

Problem 9: Converse truth

True conditional: If a figure is a square, then it is a rhombus. Is the converse always true?

Answer: **Verification:**

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Problem 10: Write conditional

Hypothesis: angle is straight. Conclusion: measure is $180°$. Write if–then.

Answer: **Verification:**

Problem 11: Inverse

If $AB = CD$, then $\overline{AB} \cong \overline{CD}$. Inverse?

Answer: **Verification:**

Problem 12: Contrapositive

If $m\angle A = 90°$, then $\angle A$ is right. Contrapositive?

Answer: **Verification:**

Problem 13: Symbols

Symbolic form of "if p then q"?

Answer: **Verification:**

Problem 14: Challenge — Law of Syllogism

(1) If a polygon is a regular hexagon, then all its sides are congruent. (2) If all of a polygon's sides are congruent, then it is equilateral. Combine using the Law of Syllogism.

Answer: **Verification:**

Problem 15: Challenge — Law of Detachment

If two angles form a linear pair, then they are supplementary. $\angle 1$ and $\angle 2$ are a linear pair, and $m\angle 1 = 112°$. Find $m\angle 2$ and name the law used.

Answer: **Verification:**

✓ Answer Key

1. n is even
2. n is divisible by 2
3. If a polygon has 3 sides, then it is a triangle
4. If an angle is not right, then it does not measure $90°$
5. If two lines intersect, then they are not parallel
6. Contrapositive
7. $x = -3$
8. When both the conditional and its converse are true
9. No
10. If an angle is straight, then its measure is $180°$
11. If $AB \neq CD$, then $\overline{AB} \ncong \overline{CD}$
12. If $\angle A$ is not right, then $m\angle A \neq 90°$
13. $p \rightarrow q$
14. If a polygon is a regular hexagon, then it is equilateral
15. $m\angle 2 = 68°$; Law of Detachment

Great work — keep practising!